

Theme: Smart Mobility

In places like Dublin's Docklands, getting around can feel harder than it should. Traffic backs up, transport options aren't always used well, and shared services, buses, e-scooters, ride-shares, can leave people waiting longer than expected. The knock-on effect is familiar: time slips away, emissions creep up, and the whole commute becomes a worse experience.

Proposed Application: DockMove

DockMove is a desktop Java Swing application that simulates and manages smart mobility services in a city setting. The point is simple: make it easier to handle transport availability and queues as conditions change, and do it in a way that's visible and fair to users.

With DockMove, users can: Register commuters. Request a mode of transport (Bus, Taxi, or E-Scooter). Enter and exit transport queues. Follow journeys while they're happening and after they're finished. Check system activity logs to see what's been going on.

Problem Being Solved

Smart city transport demand doesn't stay still, it rises and drops throughout the day. When requests and queues aren't managed properly, a few predictable issues show up fast: people wait too long, vehicles or services end up allocated badly, and users can't tell where they stand in the order of service.

DockMove addresses that by leaning on clear, structured handling through Abstract Data Types (ADTs). A Queue is used to keep allocation first-come, first-served. A Stack holds the most recent activity so the latest actions are easy to track. And an ArrayList keeps a record of all registered users along with completed trips.

How It Fits Smart Mobility

The application lines up with Smart Mobility ideas by digitally managing demand, simulating transport allocation, and pushing for fairness and better flow. The scope is simplified, but the logic resembles what you see in real tools behind ride-hailing services, smart bus scheduling, and shared mobility platforms.

Technical Approach

DockMove is built with Java and object-oriented design at the core, paired with a Swing GUI developed in NetBeans. Interfaces and ADTs shape how the system behaves. It also demonstrates key OOP concepts: inheritance and polymorphism across different transport types, encapsulation through the way objects are designed, and interfaces to define shared transport behaviour.

Conclusion

DockMove works as a scalable prototype showing how careful data structures and solid object-oriented design can improve the way urban transport services are managed. It supports the Smart Docklands Smart Mobility theme by modelling transport management that is efficient, fair, and easy for users to follow.

